

COSMOS

PRACTICAL OF JAVA PROGRAMM

AN ISO 9001:2015 CERTIFIED INSTITUTE

COSMOS

The Real Training Centre

COSMOS

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➤ Chapter -1: Introduction of Java

○ Java Datatype and variable

1. WAP to print hello world.
2. WAP to perform sum of two numbers.
3. WAP to get data like rno,name,percentage and print it.

○ Control statement and Looping statement

4. WAP to input number and check that value is zero, negative or positive.
5. Write a menu driven program to perform all arithmetic operations.
6. WAP to find factorial number of given number.
7. WAP to print cosmos computer 5 times using all three loops.

○ Array

8. WAP to input 5 array values and calculate total of array values.
9. WAP to input two matrix value 2*2. Calculate the sum of two matrix values and store in other 2*2 matrix.

➤ Chapter – 2 Java object and class

○ Class-objects

10. Write a program to create a class called "Person" with a name and age attribute. Create two instances of the "Person" class and print their name and age.
11. Write a program to create a class bank with attributes amount and three methods like credit, debit, display.

○ Constructor

12. WAP using default constructor.
13. WAP using parameter constructor.
14. WAP to pass the object as argument and display data of class(copy constructor).

- **“static” keyword**

15.WAP to demonstrate static variable,method and block.

- **“this” keyword**

16.WAP to demonstrate “this” keyword to refer current class instance.

- **“super” keyword**

17.WAP to demonstrate “super” keyword.

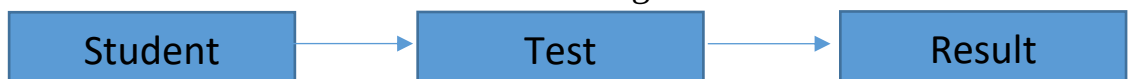
➤ **Chapter – 3 Java Inheritance and Polymorphism**

- **Inheritance**

18.Define a base class EMPLOYEE that can store salary. It has one derived class PROGRAMMER that can store bonus. Display salary and bonus of a Programmer.

19.Define base class student (int no, String name) derive class test from student (int mark1, int mark2) and derive class result from test. Consider the following figure to solve this problem.

The data member and member functions are given below:



Student class: Data members: Roll no, name

Member Functions: Getnumber(int), putnumber().

Test class: Data members: Sub1, Sub2

Member Functions: Getmarks(float,float), putmarks().

Result class: Data members: Total

Member Functions: Display()

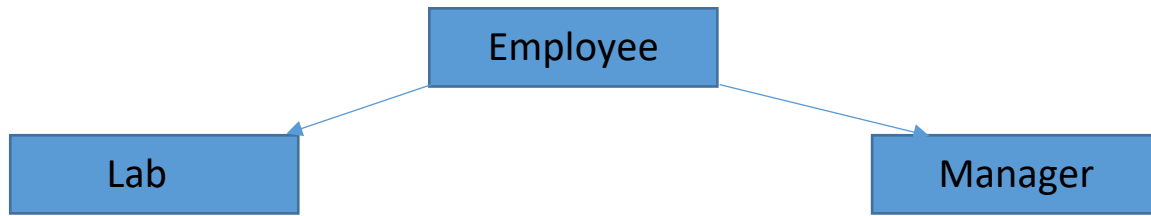
20.WAP for hierchical inheritance by considering the following figure.

Employee class: Data members: Eno,name,salary

Lab class: Member Functions: Getdata(), putdata().

Manager class: Data members: pname

Member Functions: Getm(float,float), putm().



○ **Function overloading and function overriding**

- 21.WAP to create two methods, first add() can perform addition of two numbers and second add() can perform addition of three numbers.
- 22.WAP to create class Shape with 3 methods, first area() calculate area of circle, second area() calculate area of rectangle and third area() calculate area of cube.
- 23.WAP to create class vehicle that has run(). Override this method in Bike class, and car class.
- 24.WAP to create class Bank that has getInterest(). Override this method in ICICI class, SBI class and IDBI class. Calculate maximum interest.

○ **“final” keyword**

- 25.WAP to demonstrate “final” keyword.

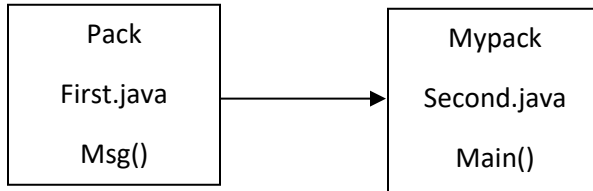
➤ **Chapter – 4 Java Abstract class, Interface and Packages**

○ **Abstract class and Interface**

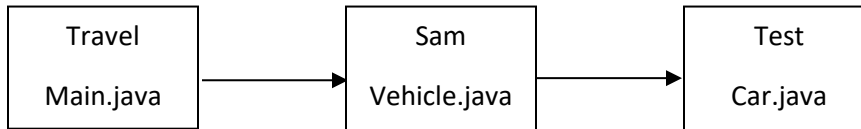
- 26.WAP to create an abstract class named “Shape” which has one abstract method. Implement this method in “Rectangle” and “Circle” classes.
- 27.WAP to create an abstract class named “Operation” which includes:
 - Data members: Number1, Number2 and Answer.
 - Abstract method: Calculation ().
 - Non – abstract method: Getdata().Implement of above data in Addition, Subtraction, Multiplication and Division classes.

○ Package

28. Create a directory structure as shown in image.



29. Create a directory structure as shown in image.



➤ Chapter – 5 Java Multithreading Programming and Exception handling.

○ Exception handling

30.WAP to illustrate Exception handling.

○ Thread and Multithread Programming

31.WAP to create thread by extending thread class and implementing runnable interface.

32.WAP to demonstrate all Thread Methods.

(sleep(),yield(),join(),getName(),setName(),isAlive()).

➤ Chapter – 6 Java String handling, File handling.

○ String class and String buffer class

33.WAP to demonstrate String class.

34.WAP to demonstrate all String function.

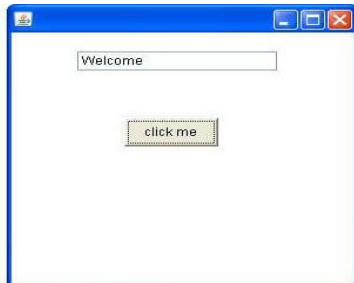
○ File Management

35.WAP to create two files named “Odd” and “Even”. Write odd and even numbers from 1-100. Read these data from file.

36.WAP to calculate mark sheet of students using serialization in java.

➤ Chapter – 8 Java AWT Class

37.WAP to demonstrate frame class.



38.WAP to demonstrate all awt control.

39.WAP to demonstrate MouseAdapter and MouseMotionAdapter class.

40.WAP to demonstrate WindowAdapter class.

41.WAP to demonstrate KeyAdapter class.